

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
10.	(a)	(i)		award (1) for any of following - ensure same distance between flame and test-tube each time - use same volume of limewater each time - use same number of moles of carbonate - ensure carbonates all have same particle size			1	1		1
		(ii)		to prevent 'suck back' / cold solution entering hot boiling tube and cracking it			1	1		1
		(iii)		thermal stability increases as you go down the group		1		1		
		(iv)		limewater would not turn cloudy because using two Bunsen burners does not increase the temperature (to over 1360 °C)			1	1		1
	(b)	(i)		moles $\text{Ba}(\text{NO}_3)_2 = 3.68 \times 10^{-3}$ (1) moles of gas = 9.20×10^{-3} (1) volume of gas = $0.225 \text{ dm}^3 = 225 \text{ cm}^3$ (1)		3		3	2	
		(ii)		student incorrect award (1) for either of following this relationship is true when temperature is in K volume would be 244 cm^3 at this temperature			1	1		